AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

الجامعة العربية المفتوحة - مصر
Arab Open University
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الامتحانات
Arab Republic of Egypt

M109
.Net Programming

Midterm Examination (MTA)
Spring Semester 2025/2026
Date: 14 / Apr. / 2026

Number of Exam Pages: 5 (including this cover sheet)	Time Allowed: 1.5 Hours (90 Mintes)

nstructions:

- Total Marks: **30 marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Part 1: Multiple Choice Questions (MCQ) [6 marks, 1 for each question]

1- What will be the output of the following code:

```
int abc = 7;
Console.Write(abc-- * --abc + " and now abc = " + abc);
```

a- 49 and abc = 5
b- 35 and abc = 6
c- 42 and abc = 6
d- 35 and now abc = 5

2- The ..... loop is a *posttest* loop, which means it performs an iteration before testing its Boolean expression.

a- while
b- do-while
c- for
d- None of the given options

3- A ..... translates a high-level language program to a separate machine program for CPU to read and execute, while ..... reads, translates, and executes the instructions of a high-level language program

a- interpreter, compiler
b- assembler, interpreter
c- assembler, compiler
d- compiler, Interpreter

4- ..... provides automatic code completion as you write programming statements by providing an array of options that make language references easily accessible.

a- EventArgs
b- PictureBox
c- Checked
d- IntelliSense

5- A named constant is a name that represents a value that cannot be changed during the program's execution, and it can be declared by the keyword .....

a- constant
b- Const
c- fixed
d- final

6- ..... allow users to select one choice from several possible choices, while ..... allow users to select multiple options.

a- Radio buttons, button
b- Picture boxes, Check buttons
c- Check boxes, List boxes
d- Radio buttons, Check boxes

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Part 2: Short Questions (MCQ) [12 marks]

Question 2: (4 Marks)

A. Rewrite the following code the ternary operator (:?) Instead of (if else) statements **(2 Marks)**.

```
if (value1 > 0) result = 4 * (value2 + 5) / 2;
else result = 4 * (value2 - 5) * 2;
```

B. What are the values of result in the following both cases: **(2 Marks)**.

- value1 = 10 and value2 = 8)
- value1 = -17 and value2 = -8)

Question 3: Trace the following code segment and specify the final output. **(4 Marks)**

```
int a = 216;
int b = a-- / 3 / 5;
int c = b * 13;
Console.WriteLine("a = " + a + ", b = " + b + " and c = " + c);
a = c-- + b;
b = a * a / 12 / 105;
c = b++ + 17;
Console.WriteLine("a = " + a + ", b = " + b + " and c = " + c);
```

Question 4: The following piece of code writes the following 2 lines in a file named "D:\M109_MTA_File.txt". Complete the missing line code. **(4 Marks)**

```
>> Welcome to Arab Open University
>> Current Semester: Spring 25-26
```

```
string filePath = .....
try {
    StreamWriter stWrt = .....
    stWrt.WriteLine.....
    stWrt.WriteLine.....
    stWrt.....
} ..... (IOException ex) { ..... }
```


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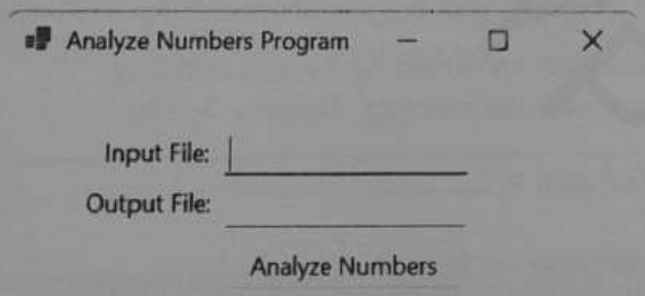
Part 3: Problem Solving Questions [12 marks]

Question 5: Write a C# program using Windows Forms that processes integer numbers as follows: **(6 Marks)**

- Read a list of integer scores (one per line) from a file named **data.txt.**
- Count how many **Odd** integers and how many **Even** integers.
- Write the results to a file named **results.txt** in the following format:
  - *Odd: [count]*
  - *Even: [count]*
- Your program must include the following controls:
  - A textbox named ***txtDataFile*** to hold the input file name
  - A textbox named ***txtResultsFile*** to hold the result data file name
  - a Button labeled ***Analyze Numbers*** and named ***btnAnalyzeNumbers*** to start the process.
- Provide a MessageBox to notify the user when the input data file is ready or if the input file is missing.

Ensure your program correctly handles file I/O, gracefully ignores any lines that are not valid integers, and provides feedback to the user (e.g., a MessageBox indicating completion or if the file was not found).

Considering the above specification, write the code of the button *btnAnalyzeNumbers* when it is clicked the numbers in the data file will be read,then do the odd/even analysis and finally write the numbers of odd and even numbers in the result file given it name in the textbox txtResultsFile. if there is no data file name is given in the textbox txtDataFile, an error message should be shown.



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Question 6: (6 Marks) Body Mass Index (BMI) is a measure of body fat and is commonly used within the health industry to determine whether the weight of adult men and women is healthy. BMI is calculated in relation to one's height using the following formulas:

BMI = weight in kilogram / (height * height) in meter

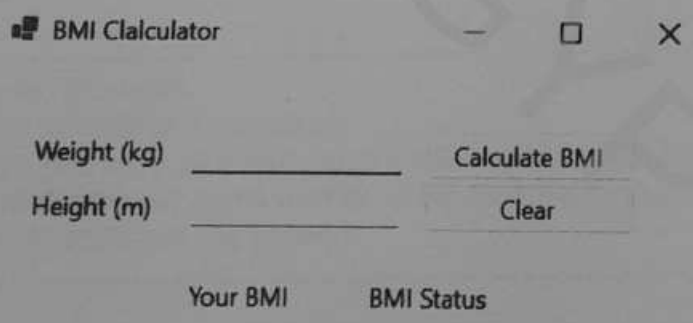
While adult BMI categories are:

- Underweight: BMI Below 18.5
- Normal Weight: BMI between 18.5 – 24.99
- Overweight: BMI between 25 – 29.99
- Obese: BMI 30.0 and above

The following form represents creating a BMI calculator in C# Windows Forms involves designing a user interface and implementing the BMI formula in the code-behind. BMI is calculated. The following controls are added to the main form Form1 using the Visual Studio designer:

- Label for "Weight (kg):" named *lblWeight* with initial text "Wight (kg):"
- TextBox for weight (in kilogram) input (name it *txtWeight*)
- Label for "Height (m):" named *lblHeight* with initial text "Height (m):"
- TextBox for height (in meter) input named txtHeight
- Button for "Calculate BMI" (name it *btnCalculateBMI*) to calculate the BMI value and status.
- Button for "Clear" (name it *btnClear*) to clear thetextboxes and update the labels to their initial values.
- Label to display the BMI value (name it *lblBMIValue*) with initial text "BMI Value:"
- Label to display the BMI status (name it *lblBMIStatus*) with initial text "BMI Status:"

Considering the above specification, write the code of the two buttons btnCalculateBMI and btnClear. If button btnCalculateBMI is clicked while there are no values in the two text boxes, an error message should be shown.



End of the questions.

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